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AMENDMENT

Vessels for commercial transactions Amendment (2009)

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Vessels for commercial transactions Amendment (2009)

Introduction

OIML R 138:2007 was approved by the CIML during its 42nd Meeting held in Shanghai. In the meantime it was decided to amend OIML R 138 to take into consideration late comments.

Amendment (2009) to OIML R 138:2007 (E)

Section 2.6

2.6 Conventional true capacity

a) For measuring container bottles and casks: Contained volume of water at a reference temperature with an uncertainty appropriate for the given purpose.

b) For capacity serving measures: Delivered volume of water at a reference temperature with an uncertainty appropriate for the given purpose.

Section 5.2.3

5.2.3 Measuring container bottles

In addition to the requirements of 5.2.1, measuring container bottles shall be marked with one of the following marks, either on the lower rim or the bottom:

— when filled to a constant level: the distance, expressed in millimeter, followed by the symbol mm from the brim to the filling level corresponding to the nominal capacity; or

— when filled to a constant ullage: the number, expressed in millilitre (ml) or centilitre (cl) not followed by the symbol mlcl^{-1} that is equal to the brim capacity.

Annex A

Annex A is an informative annex.

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INTERNATIONAL STANDARD

Vessels for commercial transactions

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Foreword

The International Organization of Legal Metrology (OIML) is a worldwide, intergovernmental organization whose primary aim is to harmonize the regulations and metrological controls applied by the national metrological services, or related organizations, of its Member States. The main categories of OIML publications are:

— **International Recommendations (OIML R)**, which are model regulations that establish the metrological characteristics required of certain measuring instruments and which specify methods and equipment for checking their conformity. OIML Member States shall implement these Recommendations to the greatest possible extent;

— **International Documents (OIML D)**, which are informative in nature and which are intended to harmonize and improve work in the field of legal metrology;

— **International Guides (OIML G)**, which are also informative in nature and which are intended to give guidelines for the application of certain requirements to legal metrology; and

— **International Basic Publications (OIML B)**, which define the operating rules of the various OIML structures and systems.

OIML Draft Recommendations, Documents and Guides are developed by Technical Committees or Subcommittees which comprise representatives from the Member States. Certain international and regional institutions

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also participate on a consultation basis. Cooperative agreements have been established between the OIML and certain institutions, such as ISO and the IEC, with the objective of avoiding contradictory requirements. Consequently, manufacturers and users of measuring instruments, test laboratories, etc. may simultaneously apply OIML publications and those of other institutions.

International Recommendations, Documents, Guides and Basic Publications are published in English (E) and translated into French (F) and are subject to periodic revision.

Additionally, the OIML publishes or participates in the publication of **Vocabularies (OIML V)** and periodically commissions legal metrology experts to write **Expert Reports (OIML E)**. Expert Reports are intended to provide information and advice, and are written solely from the viewpoint of their author, without the involvement of a Technical Committee or Subcommittee, nor that of the CIML. Thus, they do not necessarily represent the views of the OIML.

This publication — reference OIML R 138, Edition 2007 (E) — was developed by the OIML Technical Subcommittee TC 8 *Measurement of quantities of fluids*. It was approved for final publication by the International Committee of Legal Metrology in 2007 and will be submitted to the International Conference of Legal Metrology in 2008 for formal sanction. This Edition replaces the previous editions of OIML R 4 *Volumetric flasks (one mark) in glass* (Edition 1972), R 29 *Capacity serving measures* (Edition 1979), R 45 *Casks and barrels* (Edition 1980) and R 96 *Measuring container bottles* (Edition 1990).

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Vessels for commercial transactions

1 Scope and application

This Recommendation applies to volumetric containers or vessels used to measure, and when applicable to store and transport, liquids for commercial transactions.

2 Terminology and symbols

2.1 General terms

The general terms and definitions are those of the International Vocabulary of Basic and General Terms in Metrology (VIM, 1993).

2.2 Capacity serving measures

Measures which are filled as required and used for the retail sale of drinks sold by volume. They are divided into two categories: transfer measures used solely for decanting specific volumes of beverages and drinking measures used also for the consumption of specific volumes of beverages.

2.3 Measuring container bottles

Bottles intended to be filled either at constant level or at constant ullage with sufficient accuracy without the need to use an independent measuring instrument.

2.4 Casks and barrels

Containers used for commercial transactions of liquids when they are filled to their nominal volume. To simplify the text, casks and barrels are referred to hereafter by the name “casks” only.

2.5 Nominal capacity, V_n

Volume indicated on the vessel.

2.6 Conventional true capacity

For measuring container bottles and casks: contained volume of water at a reference temperature with an uncertainty appropriate for the given purpose.

For capacity serving measures: delivered volume of water at a reference temperature with an uncertainty appropriate for the given purpose. This volume is measured after prior wetting of the interior of the vessel and draining for 30 seconds.

2.7 Brim capacity, V_r

Volume that the vessel is meant to contain when filled to the brim.

2.8 Ullage (or headspace)

Empty volume remaining in a container after it is filled.

2.9 Gauge mark

Mark, generally a line that indicates the nominal capacity V_n.

2.10 Filling orifice

Opening in the cask wall defined by its section and the thickness of the cask walls.

3 Units of measurement

The units of measurement shall be the SI units.

4 Technical requirements

4.1 Nominal capacities

4.1.1

Capacity serving measures and measuring container bottles

Capacity serving measures and measuring container bottles shall have one of the nominal capacities given by the following formula:

$$V_n = x \cdot 10^n \mathrm{L} \tag{1}$$

with n = positive or negative integer, or zero
$x = 1, 1{,}5, 2, 2{,}5, 3, 4$ and 5

and comprised between the following values:

— capacity serving measures [20 … 5000] mL

— measuring container bottles [50 … 5000] mL

4.1.2 Casks

Casks shall have a nominal capacity equal to:

— a multiple of $5\,\mathrm{L}$ for $V_n \leq 100\,\mathrm{L}$

— a multiple of $50\,\mathrm{L}$ for $V_n > 100\,\mathrm{L}$

4.1.3 Exceptions

When justified by local traditions or by special requirements, national regulations may allow additional values for nominal capacities or units (e.g. $0{,}33\,\mathrm{L}$).

4.2 Material

The vessels shall be sufficiently robust to withstand normal usage without distortions.

For capacity serving measures and measuring container bottles, the materials used in construction should be sufficiently tough, rigid, non porous and stable to maintain accuracy during use.

Casks may be manufactured in any material with strength, rigidity and toughness adequate for the intended use (e.g. solid wood, plywood, metal). National regulations may stipulate that the interior of certain casks must be covered with a protective permanent coating, compatible with the material used for construction and the liquids to be contained.

Vessels used only once, such as thin plastic or cardboard drinking capacities, must remain accurate during nominal filling, transportation and discharge functions.

4.3 Construction

The total volume of vessels intended to contain liquids under pressure shall not increase more than 0.5 % when subjected to an internal pressure of $100\,\mathrm{kPa}$ maintained for 48 hours.

The position of the filling orifice shall allow for complete filling of the casks.

4.4 Resolution if gauge mark

Vessels with gauge marks shall be so designed that at the nominal capacity, a variation in the contents equal to the maximum permissible error causes a change in the liquid level of at least $2\,\mathrm{mm}$.

4.5 Vertical position

Vessels with gauge marks and measuring container bottles shall have a base wide enough to allow a vertical, stable position when resting on a flat horizontal surface. If not, a holder must be provided with the vessel.

4.6 Filling requirements

4.6.1 Vessels with gauge marks

“Filled to the gauge mark” is when the lowest part of the meniscus formed by the liquid is tangential to the upper edge of the gauge mark.

4.6.2 Measuring container bottles

Depending on the specifications of the manufacturer, the surface of the liquid shall either be at the specified distance from the brim or have the specified ullage at the top.

4.6.3 Casks

Casks shall be filled to the lower edge of the filling orifice.

5 Metrological requirements

5.1 Maximum permissible errors

5.1.1 Capacity serving measures

Table 1 — Capacity serving measures

		Maximum permissible error (MPE)	
		Nominal capacity V_n	**Brim capacity V_r**
Transfer measures	$< 100\,\mathrm{mL}$	$\pm 2\,\mathrm{mL}$	0
			$+4\,\mathrm{mL}$
	$\geq 100\,\mathrm{mL}$	$\pm 3\%$ of V_n	0
			$+6\%$ of V_r
Drinking measures	$< 200\,\mathrm{mL}$	$\pm 5\%$ of V_n	0
			$+10\%$ of V_r
	$\geq 200\,\mathrm{mL}$	$\pm(5\,\mathrm{mL} + 2{,}5\%$ of $V_n)$	0
			$+10\,\mathrm{mL} + 5\%$ of V_r

5.1.2 Measuring container bottles

Table 2 — Measuring container bottles

Nominal capacity V_n or Brim capacity V_r in mL	**Maximum permissible error (MPE)**	
	in % of V_n or V_r	**in mL**
from 50 to 100	-	± 3
from 100 to 200	± 3	-
from 200 to 300	-	± 6
from 300 to 500	± 2	-
from 500 to 1000	-	± 10

Table 2 *(continued)*

Nominal capacity V_n or Brim capacity V_r in mL	Maximum permissible error (MPE)	
	in % of V_n or V_r	in mL
from 1000 to 5000	± 1	-

5.1.3 Casks

$\pm 0{,}5\%$ but not less than $0{,}10\,\mathrm{L}$ for casks made of metal which shall be designated by the letter “A” affixed to the vessel.

$\pm 1{,}0\%$ but not less than $0{,}15\,\mathrm{L}$ for casks made of other materials which shall be designated by the letter “B” affixed to the vessel.

5.1.4 General

Systematic exploitation of the maximum permissible errors is prohibited.

5.2 Markings

5.2.1 Nominal capacity

Every vessel for commercial transactions shall bear a clearly visible, easily readable and permanent indication of the nominal capacity V_n. Markings shall be in SI units.

5.2.2 Capacity serving measures

In addition to the requirements of 5.2.1, capacity serving measures shall bear at least one filling mark. All filling marks shall be clearly visible and permanently marked. They must be at least $10\,\mathrm{mm}$ long and in a horizontal plane when the measure stands on a horizontal flat surface.

On drinking measures with a capacity of more than 50 mL, the filling mark is at least $10\,\mathrm{mm}$ from the brim.

For foam building liquids the distance between the upper gauge mark and the brim shall be sufficient to allow for a complete filling to the mark but not less than $20\,\mathrm{mm}$.

Exception: if the capacity of a serving or drinking measure is defined by its brim capacity, the requirement of 5.2.2 does not have to be fulfilled.

5.2.3 Measuring container bottles

In addition to the requirements of 5.2.1, measuring container bottles shall be marked with one of the following marks, either on the lower rim or the bottom:

— when filled to a constant level: the distance, expressed in mL followed by the symbol mm from the brim to the filling level corresponding to the nominal capacity; or

— when filled to a constant ullage: the number, expressed in mL followed by the symbol mL that is equal to the brim capacity.

5.2.4 Compliance with the metrological specifications

Every vessel for commercial transactions shall bear a mark identifying the company responsible for compliance with the metrological specifications applicable to the vessel.

5.2.5 Reference temperature

If the National Metrology Authority decides on a reference temperature other than $20\,^{\circ}\mathrm{C}$, this temperature must be indicated next to every volume indication affixed to the vessel.

6 Metrological controls

6.1 Type approval

The vessels for commercial transactions are not subject to type approval.

6.2 Initial verification

The conformity of vessels used for commercial transactions to the prescribed specifications shall be controlled by the competent authority using a suitable statistical test. The controls shall preferably take place at the manufacturer's plant or at the importer's warehouse.

The competent authority shall use a suitable test to perform the controls. An example is given in Annex A. In cases where a small number of vessels are concerned, the control may be replaced by individual initial verification.

Casks may be presented to the competent authority for legal metrology without an indication of the nominal capacity. In this case, the authority shall determine the conventional true capacity at a reference temperature when the container is filled according to the specifications and shall affix the value of the nominal capacity and the official mark of verification.

Capacity serving measures may be presented to the competent Legal Metrology Authority without marking. In this case, the authority shall affix the gauge mark, the value of the nominal capacity and the official verification mark.

6.3 Subsequent verification or control

Vessels made of glass are not subject to subsequent verification.

7 Measuring instruments used for official controls

Measuring instruments used for official controls of vessels for commercial transactions shall be traceable to internationally supported realizations of the SI Units and have an uncertainty not greater than one-third of the maximum permissible error.

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